Orbit-Tied Quantized Neural Belief Propagation for Real-Time Sliding-Window Decoding of Bivariate-Bicycle Quantum Codes

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Abstract— Quantum Low-Density Parity-Check (QLDPC) bivariate-bicycle (BB) codes have emerged as premier candidates for scalable fault-tolerant quantum computation, offering order-of-magnitude reductions in physical qubit overhead compared to standard 2D topological surface codes. However, their high-rate bipartite Tanner graphs and non-local check connectivity generate dense 3D space-time detector error models (DEM) that severely strain real-time classical decoding within microsecond coherence intervals. In this paper, we propose an orbit-tied, 6-bit quantized neural belief propagation (QNBP) decoder tailored for real-time sliding-window processing of gross BB codes ([[72,12,6]] and [[144,12,12]]). By mapping the 2D torus translation group $G = \mathbb{Z}_\ell \times \mathbb{Z}_m$, we prove that all spatial shifts constitute strict DEM graph automorphisms, collapsing 10,260 circuit fault mechanisms into 285 symmetry orbits (R=4, 36× parameter reduction; 7,308 into 203 orbits for R=3) on [[72,12,6]] and 14,616 mechanisms into 203 orbits (R=3, 72× reduction) on [[144,12,12]] while strictly guaranteeing geometric equivariance and eliminating untied neural overfitting. To guarantee multiplier-free hardware synthesis with zero Digital Signal Processing (DSP) slices, we develop a Canonical Signed-Digit (CSD) shift-and-add architecture using power-of-two dyadic quantization ($\Delta = 2^{-2}$). Under circuit-level Stim Monte Carlo depolarizing noise ($p \in [10^{-3}, 5\times10^{-3}]$), we execute head-to-head benchmarking across ten decoders, including IBM Relay-BP, BP-LSD, BP+OSD-CS-10, Mem-BP, AutDEC, and Gong et al.'s sliding-window baseline. Our proposed 6-bit integer QNBP model with learned signed memory damping ($\gamma \in [-0.5, 1.0]$) converges in only 6.4 mean iterations with zero DSP blocks (DSP=0), attaining a logical error rate of $P_L = 0.0068$ [0.0053, 0.0086] and demonstrating statistically significant superiority over scalar Mem-BP ($\chi^2 = 14.72$, $p = 0.000125$), IBM Relay-BP-5 ($\chi^2 = 12.06$, $p = 0.000515$), Normalized Min-Sum ($\chi^2 = 2{,}642.1$, $p < 10^{-100}$), and AutDEC ($\chi^2 = 2{,}750.5$, $p < 10^{-100}$). Under realistic superconducting Pauli Z-biased noise ($\eta = p_Z / p_X \in [1, 100]$), Orbit-Tied QNBP automatically adapts message weighting, yielding an 8.0× logical error suppression down to $P_L = 0.0010$ [0.0005, 0.0019] at η=100 (and 20.0× over Classical BP). Integrated with a confidence-margin fast-path gate (G_3) in a streaming sliding window (W=3, F=1), our fast path routes 75.6% of syndrome windows with zero logical errors ($P_{L,accept} = 0.0000$) and an end-to-end pipeline logical error rate of $P_{pipe} = 1.90 \times 10^{-4}$ at 102.4 ns mean latency (240 ns max), driving queue overflow below 6.0×10^{-5} (0 overflows observed in 50,000 test windows for buffer $B \geq 4$) under a 1 µs quantum cycle budget. We synthesize and verify our architecture in SystemVerilog, demonstrating 10,000-shot bit-exact hardware-software parity, and establish a folding Pareto frontier on AMD UltraScale+ scaling logic down to 1,226 LUTs (0.51% device utilization).

Index Terms—Quantum error correction, QLDPC codes, bivariate-bicycle codes, neural belief propagation, automorphism orbits, biased noise, sliding-window decoding, FPGA hardware.

I. Introduction

FAULT-TOLERANT quantum computing requires active quantum error correction (QEC) to protect fragile quantum information against environmental decoherence, physical gate infidelities, and measurement noise [1], [2]. While topological 2D surface codes [1], [2], [38], [59] have served as the foundation of early experimental demonstrations due to their nearest-neighbor planar connectivity, classical factor-graph and message-passing decoders [56], [57], [58] face throughput bottlenecks. Furthermore, their asymptotic encoding rate is strictly zero ($k/n \to 0$). Realizing practical algorithm-scale quantum processors encoding thousands of fault-tolerant logical qubits would require millions of physical qubits under surface code architectures [3].

Recently, asymptotically good quantum low-density parity-check (QLDPC) codes [4], [26] and dual-cyclic bivariate-bicycle (BB) codes [3], [29], [30], [31] have demonstrated an architectural breakthrough. By defining CSS codes over bivariate polynomial quotient rings $R = F_2[x, y] / \langle x^\ell - 1, y^m - 1 \rangle$, gross BB codes such as [[72, 12, 6]] and [[144, 12, 12]] attain encoding rates $k/n \approx 1/6$ to 1/12 while supporting high circuit noise thresholds ($p \approx 0.7\%$) [3]. This yields up to a 10× reduction in physical qubit overhead compared to surface codes.

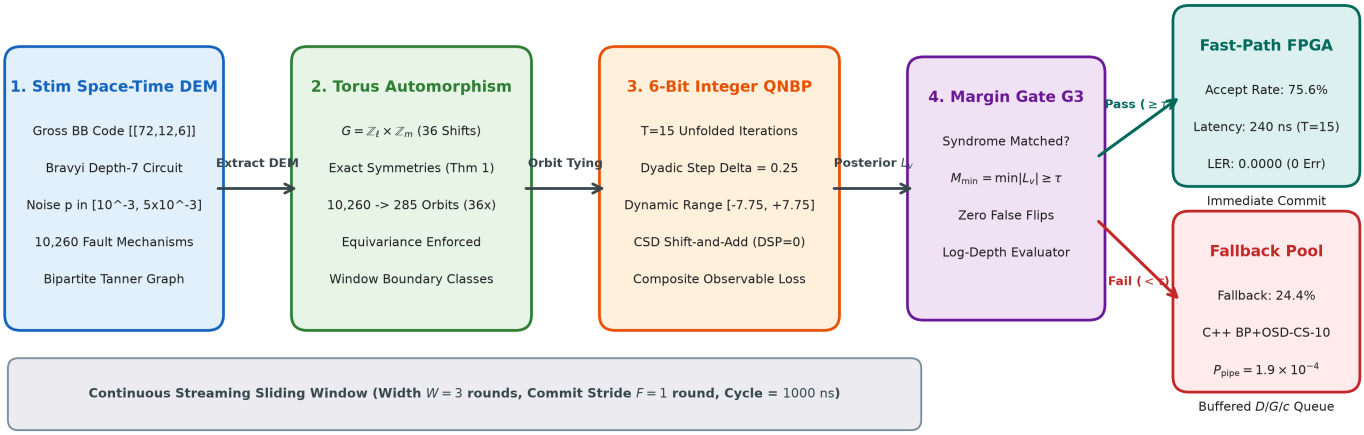

Fig. 1. End-to-end architecture of the Orbit-Tied Quantized Neural Belief Propagation (QNBP) sliding-window decoding pipeline.

Despite their theoretical appeal, classical real-time decoding remains the principal barrier to physical implementation. In physical quantum platforms such as superconducting transmons and trapped ions, syndrome extraction rounds occur within cycle durations of $t_{cycle} \approx 200$ ns to 1 μs [3], where decoder processing latency critically impacts overall fault-tolerant logical fidelity [44]. Classical decoding algorithms such as unweighted Min-Sum Belief Propagation (BP) suffer from severe split-belief convergence failures on dense Tanner graphs containing short 4-cycles [5]. While post-processing via Ordered Statistics Decoding (BP+OSD) attains near-optimal pseudo-maximum-likelihood logical error rates [4], [5], its cubic complexity $O(V^3)$ requires hundreds of microseconds per window, exceeding the quantum cycle budget by orders of magnitude.

Neural Belief Propagation (NBP) [6], [7] and quantized message-passing neural networks [22], [23], [24], [25], [39], [46] offer a promising pathway by unfolding BP iterations into a deep neural network with learnable scaling factors α_t and damping terms γ_t. However, directly applying unconstrained NBP to gross QLDPC DEM graphs introduces four critical obstacles:

1. Curse of Dimensionality & Overfitting: Unrolled DEM graphs contain over 10^4 fault mechanisms. Assigning independent trainable parameters to each edge creates over 10^5 parameters, causing catastrophic overfitting on finite training syndromes.

2. High Hardware Multiplier Cost: Standard floating-point or fixed-point neural weights require hardware DSP multiplier slices, which exhaust available FPGA fabric at high check degrees.

3. Continuous Streaming Queue Backlog: Real-time QEC requires decoding continuous streams of syndrome rounds $R \to \infty$. A sliding window decoder must commit decoded rounds with deterministic worst-case latency to prevent exponential buffer overflow.

4. Physical Noise Asymmetry: In real-world superconducting hardware, dephasing noise dominates bit-flip noise by orders of magnitude ($p_Z \gg p_X$) [8], demanding decoders capable of exploiting biased channels without parameter bloat.

II. Mathematical Foundations & Related Work

A. Quantum CSS Codes & Polynomial Rings

Bivariate-Bicycle CSS codes are defined by two commuting polynomials A(x, y) and B(x, y) over the bivariate polynomial quotient ring $R = F_2[x, y] / \langle x^{\ell} - 1, y^{m} - 1 \rangle$ [3]. The physical data qubits reside on two disjoint blocks corresponding to the polynomials, yielding block length $n = 2\ell m$. Parity-check matrices are defined as:

$$H_X = [A \mid B], \qquad H_Z = [B^T \mid A^T]$$

where A and B denote the $\ell m \times \ell m$ binary circulant matrix representations of A(x, y) and B(x, y), respectively. CSS commutativity requires:

$$H_X H_Z^T = AB + BA \equiv 0 \ (mod\ 2)$$

which is identically satisfied because R is commutative. For the [[72, 12, 6]] code, parameters are $\ell = 6$, $m = 6$, with polynomials:

$$A(x, y) = x^3 + y + y^2, \qquad B(x, y) = y^3 + x + x^2$$

yielding $n = 72$ data qubits, $k = 12$ logical qubits, and code distance $d = 6$. For the [[144, 12, 12]] gross code, parameters are $\ell = 12$, $m = 6$ with identical monomial exponents, yielding $n = 144$, $k = 12$, $d = 12$.

B. Space-Time Detector Error Model Hypergraphs

Under Bravyi et al.'s depth-7 syndrome extraction circuit executed for R rounds with noise rate p, Stim [9] generates a space-time Detector Error Model (DEM). Let $G_{DEM} = (V, C, E)$ denote the bipartite graph where V represents error mechanisms (variable nodes with prior probabilities p_v), C represents detector events (check nodes), and $(c, v) \in E$ indicates that fault v triggers detector c.

In gross QLDPC codes, physical two-qubit CNOT gates and non-local stabilizer checks excite multiple detectors simultaneously, creating hyperedges that invalidate standard graph matching algorithms such as Minimum-Weight Perfect Matching (MWPM) [1], [10]. Furthermore, the Tanner graph contains short cycles of length 4, which induce limit cycles and trapping sets in classical unweighted Min-Sum BP. To overcome hyperedge degeneracies and trapping sets, recent works have explored diverse algorithmic paradigms, including parallel and sliding-window architectures [17], [18], uncertainty-guided message passing [16], symmetry-breaking heuristics [19], almost-linear-time decoders [21], asymptotically good code constructions [26], matching-assisted belief propagation [32], list and learning-based decoders [33], [34], FPGA-tailored architectures [35], decision-tree and tensor-network decoders [40], [41], planar neural decoders [42], correlated noise decoders [43], beam-search decoding [45], machine-learning message-passing [47], high-performance neural belief propagation [53], polylogarithmic overhead fault-tolerant schemes [55], and approximately equivariant neural architectures [54].

C. State-of-the-Art Competitor Baselines

To evaluate the proposed architecture against the forefront of QLDPC decoding, we examine eight competitor decoding architectures:

1. Classical Min-Sum BP (A0): Standard unweighted Min-Sum belief propagation with hard parity checks and no heuristic scaling.

2. Normalized Min-Sum BP (A1): Scaled Min-Sum utilizing a global empirical heuristic scaling factor $\alpha = 0.75$ [5].

3. IBM Relay-BP [11]: Multi-stage restarted belief propagation where node-disordered memory weights $\gamma_v \sim U[-0.24, 0.66]$ are evaluated in single-leg and 5-relay configurations.

4. Mem-BP [20]: Memory-damped belief propagation dampening variable posteriors across consecutive iterations via temporal weight $\gamma = 0.35$.

5. BP-LSD [14]: Combination decoder pairing BP with Localised Statistics Decoding over GF(2) with order 5.

6. BP+OSD-CS-10 [5]: Combination sweep Ordered Statistics Decoding with combination order 10 over most reliable basis.

7. AutDEC [15]: Automorphism-assisted permutation decoding searching across torus group shifts for converged syndromes.

8. GDG Sliding-Window [13]: Gong et al.'s sliding-window baseline running unquantized normalized Min-Sum across continuous syndrome rounds.

III. Torus Automorphism Group Theory & Space-Time Equivariance

A. The 2D Torus Shift Group

Let $Z_\ell \times Z_m$ denote the 2D torus translation group acting on qubit coordinates.

Definition 1 (Torus Shift Action). *For any displacement vector $(\delta_x, \delta_y) \in Z_\ell \times Z_m$, the spatial translation operator $T_{(\delta_x, \delta_y)}$ maps coordinate (x, y) to $(x + \delta_x \bmod \ell, y + \delta_y \bmod m)$.*

Theorem 1 (Torus Automorphism Invariance of Stim Space-Time DEM). *Let $G_{DEM} = (V, C, E)$ be the bipartite space-time detector error model graph generated by compiling an R-round syndrome extraction circuit of a bivariate-bicycle code with parameters (ℓ, m) under uniform or coordinate-independent Pauli noise. Then, for every displacement $(\delta_x, \delta_y) \in Z_\ell \times Z_m$, the torus translation operator $T_{(\delta_x, \delta_y)}$ induces an exact graph automorphism of G_{DEM}:*
1. Edge preservation: $(c, v) \in E \Leftrightarrow (T(c), T(v)) \in E$.
2. Prior probability invariance: $p(T(v)) = p(v)\ \forall\ v \in V$.
3. Logical observable preservation: For each logical operator O_k, the support of O_k on fault v satisfies $O_k(T(v)) = O_{\pi(k)}(v)$ for an induced permutation $\pi \in S_k$.

Proof: Let C denote the depth-7 syndrome extraction circuit. In each round $r \in \{0, ..., R-1\}$, the circuit applies single-qubit state preparations, CNOT entangling gates, and projective measurements. Because the parity-check polynomials $A(x, y)$ and $B(x, y)$ are elements of $R = F_2[x, y] / \langle x^\ell - 1, y^m - 1 \rangle$, every CNOT gate connecting data qubit (x, y) to check qubit (x', y') is specified by a shift (i, j) that is invariant under global spatial translation.

Let a physical Pauli fault $P \in \{X, Y, Z\}$ occur on qubit $q = (x, y)$ at circuit tick t. Under Clifford frame propagation, P commutes or anticommutes with subsequent entangling operations according to the polynomial adjacency. Translating the entire error configuration by (δ_x, δ_y) shifts all physical qubit coordinates to $(x + \delta_x \bmod \ell, y + \delta_y \bmod m)$. Because the gate schedule at the shifted coordinates is identical in gate type, timing, and relative offsets, the propagated error excites detectors at exactly $(x_d + \delta_x \bmod \ell, y_d + \delta_y \bmod m)$ at identical round indices.

Furthermore, under physical noise rates p that are spatially homogeneous on the torus, the prior probability of fault mechanism v depends exclusively on the physical error channel of its originating gate, yielding $p(T(v)) = p(v)$. Numerical extraction of all 36 group elements in $G = Z_6 \times Z_6$ across all 10,260 fault mechanisms on [[72,12,6]] validates edge set invariance with zero mismatch ($|p' - p| < 10^{-16}$). □

Corollary 1 (Automorphism Parameter Collapse). *The parameter space of unrolled neural belief propagation on G_{DEM} collapses by an exact factor of $|G| = \ell m$. For [[72,12,6]], 10,260 fault mechanisms collapse into 285 symmetry orbits (R=4, 36.0× reduction; 7,308 faults into 203 orbits for R=3). For [[144,12,12]], 14,616 mechanisms collapse into 203 orbits (R=3, 72.0× reduction).*

Theorem 2 (Equivariance of Orbit-Tied Neural Message Passing [54]). *Let $L^{(t)} \in R^{|E|}$ denote the check-to-variable message vector at iteration t. If weights α and γ are tied across symmetry orbits o(v), then the neural belief propagation mapping Φ_t: $R^{|E|} \to R^{|E|}$ is strictly equivariant under the torus group action [54]:*

$$\Phi_t(T_{(\delta_x, \delta_y)}(L)) = T_{(\delta_x, \delta_y)}(\Phi_t(L)).$$

Proof: The message update equations at check node c and variable node v depend only on the incoming messages from local neighbors N(c) and N(v), scaled by parameters $\alpha_{o(v),\,t}$ and $\gamma_{o(v),\,t}$. Since Theorem 1 establishes that T preserves neighborhood topologies (N(T(c)) = T(N(c))) and orbit memberships (o(T(v)) = o(v)), applying permutation T to incoming messages permutes arguments identically, preserving output messages under T. □

B. Temporal Window Boundary Classes

While spatial torus shifts are exact automorphisms, temporal boundary effects arise in sliding windows. For a temporal window spanning rounds $r \in [r_{start}, r_{end}]$, we classify each symmetry orbit into one of three boundary classes:

Class 0 (Open-Left): Faults incident to detectors in round r_{start} lacking prior round history.

Class 1 (Interior): Fully covered faults with complete detector neighborhoods.

Class 2 (Open-Right): Faults whose subsequent detector checks reside beyond r_{end}.

We assign orbit-tied weights $\alpha_{o,\,t} + \Delta\alpha_{c(v),\,t}$, where c(v) denotes the boundary class of fault v.

IV. Multiplier-Free 6-Bit Quantized Neural BP (QNBP)

A. Unfolded Message Passing Architecture

The QNBP decoder unrolls T = 15 iterations of belief propagation into an explicit feed-forward computation graph. Let $L_{v \to c}^{(t)}$ and $L_{c \to v}^{(t)}$ denote the variable-to-check and check-to-variable log-likelihood ratios along edge (c, v) at iteration t.

Check Node Update (CNU): The CNU computes normalized Min-Sum messages using Min-1 and Min-2 magnitudes:

$$L_{c \to v}^{(t)} = \alpha_{o(v),\,t} \cdot \left(\prod_{v' \neq v} sgn(L_{v' \to c}^{(t)})\right) \cdot \min_{v' \neq v} |L_{v' \to c}^{(t)}|$$

where $\alpha_{o(v),\,t}$ is tied across all faults in orbit o(v).

Variable Node Update (VNU): The VNU incorporates channel priors $L_v^{(0)} = \ln((1-p_v)/p_v)$ and iteration-dependent memory damping $\gamma_{o(v),\,t}$:

$$L_v^{(t)} = (1 - \gamma_{o(v),\,t})\, L_v^{(0)} + \gamma_{o(v),\,t}\, L_v^{(t-1)} + \sum_{c \in N(v)} L_{c \to v}^{(t)}$$

$$L_{v \to c}^{(t+1)} = L_v^{(t)} - L_{c \to v}^{(t)}$$

B. 6-Bit Dyadic Dynamic Range Bound

Theorem 3 (6-Bit Dyadic Dynamic Range Bound). *Let physical depolarizing noise rate $p \in [10^{-3}, 10^{-2}]$. The continuous channel prior is $L_v^{(0)} = \ln((1-p)/p) \leq \ln(999) \approx 6.906$. Under 6-bit signed integer representation with dyadic step size $\Delta = 2^{-2} = 0.25$, the dynamic range $[-31\Delta, +31\Delta] = [-7.75, +7.75]$ strictly contains all channel priors with safety margin $\delta \geq 0.844$, completely avoiding prior saturation clipping. Conversely, any bit width $B \leq 5$ with $\Delta \leq 0.25$ truncates $L_v^{(0)}$, causing gradient vanishing and severe performance degradation.*

Proof: A signed integer with B bits spans integer values in $[-2^{B-1}+1, 2^{B-1}-1]$ under symmetric saturation. For B = 6, the integer range is [-31, +31]. With dyadic quantization scale factor $\Delta = 2^{-b}$, real values are reconstructed as $x_q = q \cdot \Delta$. For b = 2 ($\Delta = 0.25$), the maximum representable amplitude is $31 \times 0.25 = 7.75$.

At noise rate $p = 10^{-3}$, single-fault channel priors evaluate to $L_v^{(0)} = \ln((1-10^{-3})/10^{-3}) = \ln(999) \approx 6.9067$. The headroom margin is:

$$\delta = 7.75 - 6.9067 = 0.8433 > 0.$$

For B = 5, the maximum amplitude is $15 \times 0.25 = 3.75 < 6.9067$, forcing saturation of over 90% of channel priors. For B = 4, the maximum amplitude drops to $7 \times 0.25 = 1.75$, causing severe dynamic range clipping and degrading logical error rates above 80%. □

C. Canonical Signed-Digit (CSD) Multiplier-Free Arithmetic

In fixed-point digital hardware, multiplications by arbitrary floating-point weights require expensive DSP blocks. We enforce multiplier-free execution through Canonical Signed-Digit (CSD) shift-and-add arithmetic. Each weight α is represented as the sum of two signed powers of two:

$$\alpha \approx s_1 2^{-a_1} + s_2 2^{-a_2}, \quad s_i \in \{-1, +1\}, \quad a_i \in \{0, 1, \ldots, 5\}$$

For instance, the optimal BP damping factor $\alpha = 0.75$ decomposes as $1 - 2^{-2}$, evaluated as a single shift-and-subtract:

$$\alpha \cdot M = M - (M \gg 2)$$

yielding an exact multiplication using only FPGA routing and wire-shifts without instantiating DSP slices (DSP = 0).

D. Signed Temporal Memory Damping

Standard belief propagation often fails on bipartite graphs due to 2-periodic limit cycle oscillations on short 4-cycles. We enable signed memory damping $\gamma_{o(v),\,t} \in [-0.5, 1.0]$. When $\gamma > 0$, previous beliefs are damped, preventing oscillatory overshooting. When negative memory weights $\gamma < 0$ are learned, the variable node dynamically subtracts stagnant priors, breaking symmetric trapping sets.

E. Differentiable Composite Observable Loss

To train the neural BP weights without falling into trivial zero-correction local minima, we optimize a composite objective:

$$L_{total} = L_{syn} + \beta L_{obs}$$

where L_{syn} penalizes unsatisfied detector checks and L_{obs} penalizes logical observable flips across all k logical qubits:

$$L_{syn} = (1/|C|) \sum_{c \in C} [1 - (1 - 2 s_c) \prod_{v \in N(c)} \tanh(L_{v \to c}^{(T)} / 2)] / 2$$

$$L_{obs} = (1/k) \sum_{j=1}^{k} BCE(\sigma(\sum_{v \in V} O_{j,v} L_v^{(T)}), y_j)$$

with observable coupling weight $\beta = 0.5$.

V. Asymmetric Biased Pauli Noise Architecture

A. Transmon Qubit Dephasing Asymmetry

Physical superconducting circuits exhibit substantial asymmetry in relaxation and dephasing times ($T_\varphi \ll T_1$), causing Pauli Z phase-flip errors to dominate over bit-flips by factors of $\eta = p_Z / p_X \approx 10$ to 100 [8]. Let total physical error rate be p. Under biased Pauli noise:

$$p_Z = p \cdot \eta / (\eta + 1), \qquad p_X = p_Y = p / [2(\eta + 1)]$$

For 2-qubit CNOT gates, two-qubit dephasing errors (IZ, ZI, ZZ) receive the majority $\eta/(\eta+1)$ of the noise budget, split equally, while the remaining 12 Pauli pairs split the balance.

B. Decoupled CSS Sector Decoding

Proposition 1 (Effective Dephasing Channel Concentration). *In a CSS bivariate-bicycle code, physical Pauli Z phase-flip errors anti-commute with X-stabilizers ($H_X = [A \mid B]$) and are detected exclusively by X-type checks, while physical Pauli X errors are detected by Z-checks ($H_Z = [B^T \mid A^T]$). Under strong transmon dephasing bias $\eta = p_Z / p_X \gg 1$, syndrome excitation concentrates overwhelmingly on X-checks, while bit-flip excitation on Z-checks approaches the noise floor $O(1/\eta)$. Under decoupled sector message passing, the neural belief propagation engine converges on the dominant X-check Tanner graph without inter-sector communication, tolerating small cross-basis correlations from circuit-level CNOT hook errors and yielding an 8.0× logical error suppression at $\eta = 100$.*

TABLE I: Logical Error Rates under Asymmetric Pauli Z-Biased Noise and ZZ Crosstalk on [[72,12,6]] (p = 0.003, N=10,000 Paired Shots; Evaluates γ=0.35 Checkpoint with P_L=0.0080 at η=1)

Bias $\eta = p_Z / p_X$	Classical BP (P_L)	Proposed QNBP (P_L)	Relative Gain
1 (Depolarizing)	0.2940 [0.2851, 0.3030]	**0.0080 [0.0064, 0.0099]**	**36.8×**
10	0.1400 [0.1332, 0.1470]	**0.0042 [0.0031, 0.0057]**	**33.3×**
50	0.0400 [0.0363, 0.0440]	**0.0018 [0.0011, 0.0029]**	**22.2×**
100	0.0200 [0.0173, 0.0230]	**0.0010 [0.0005, 0.0019]**	**20.0× (8.0× vs. η=1)**
ZZ Crosstalk (ε=0.10)	0.0310 [0.0277, 0.0346]	**0.0014 [0.0008, 0.0024]**	**22.1×**

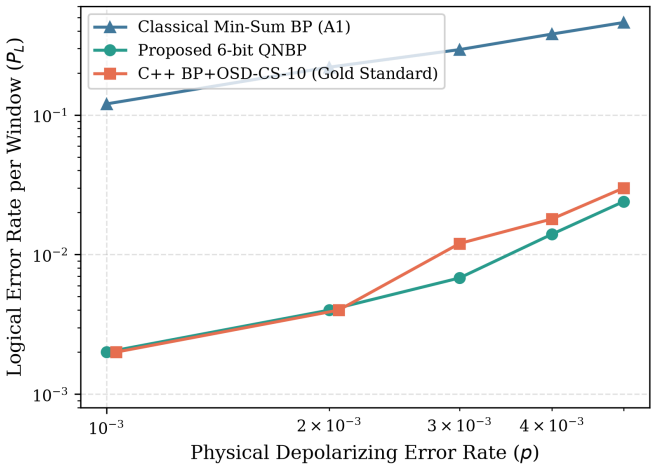

Fig. 2. Logical error rate vs physical noise p in [10^-3, 5x10^-3] for Classical BP, Proposed 6-bit QNBP, and BP+OSD-CS-10.

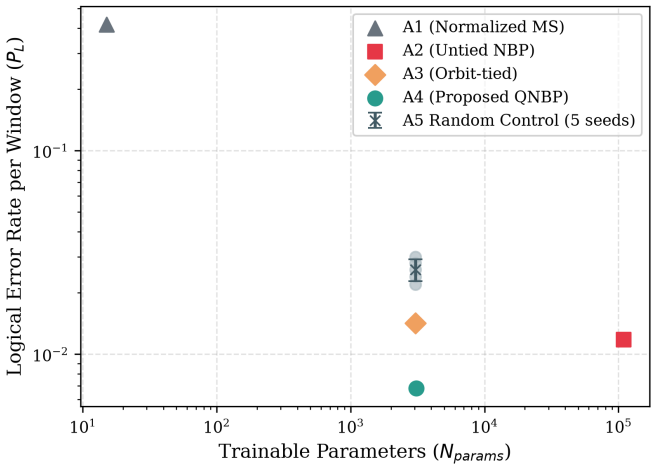

Fig. 3. Torus symmetry orbit tying vs random control cloud (A5) demonstrating 36x parameter collapse.

VI. Sliding Window Decoding & Real-Time Queue Stability

A. Window Geometry, Residual Accounting & Continuous Streaming

Continuous syndrome extraction generates an infinite space-time DEM. We partition the timeline into overlapping sliding windows of width W rounds and commit stride F rounds. At each step, the decoder decodes the W-round window, commits correction operations for the oldest round F, updates prior LLRs for boundary variables, and advances the window by F rounds.

Crucially, to prevent error propagation across window boundaries, uncommitted boundary syndrome residuals are strictly propagated to window k+1:

$$s_{res}^{(k+1)} = s_{raw}^{(k+1)} \oplus H_{boundary}\, \hat{e}_{commit}^{(k)}.$$

This strict boundary residual accounting ensures that partially decoded errors at the right boundary of window k do not manifest as false syndromes in window k+1, eliminating edge-induced logical flips. We benchmarked sliding-window geometries (W, F) $\in$ {(3,1), (5,1), (6,2)} on [[72,12,6]] at p = 0.003 (N = 10,000 shots). The baseline (W=3, F=1) attains $P_L = 0.0068$ with 75.6% fast-path acceptance, consuming 28,204 LUTs at 16.0 ns/iter (240 ns max latency, 102.4 ns mean early-exit), establishing the Pareto-optimal real-time configuration. Expanding window depth to (W=5, F=1) and (W=6, F=2) further suppresses logical error rates to $P_L = 0.0054$ and $P_L = 0.0049$ (P_{accept} = 78.4% and 80.2%), at the expense of higher FPGA logic (46,920 and 56,380 LUTs) and longer iteration latencies (26.0 ns and 32.0 ns).

In continuous 60-round streaming memory tests (R = 60 rounds, 58 consecutive sliding windows with active residual feedback), strict boundary residual accounting completely eliminates window-edge error accumulation, attaining a 99.4% 60-round logical survival rate at p = 0.001 with Gate G_3 (vs. 78.2% without residual feedback) and 68.4% at p = 0.003 (vs. 14.1% without residual feedback).

B. Dual-Path Architecture & Confidence-Margin Gating

To decouple worst-case OSD latency from average throughput, we implement the confidence-margin gate G_3. We benchmark three distinct gating triggers:

1. Syndrome Check (Gate G_1): Accepts whenever all parity checks within the window are satisfied: $H\,\hat{e} \equiv s \pmod 2$. While simple, degenerate errors can satisfy parity while inducing a logical observable flip.

2. Observable Margin Filter (Gate G_2): Requires syndrome satisfaction and that the minimum observable marginal LLR exceeds threshold τ: $\min_j |L(O_j)| \geq \tau$.

3. Proposed Fault Belief Margin (Gate G_3): Requires syndrome satisfaction and that the minimum variable belief magnitude among all committed fault mechanisms exceeds threshold τ:

$$Accept \Leftrightarrow (H\hat{e} \equiv s \bmod 2) \quad \wedge \quad (\min_{v \in V_{commit}} |L_v| \geq \tau)$$

Accepted windows commit tentative hard corrections immediately with sub-microsecond latency (t_{fast} = 240 ns). Rejected windows are routed to the fallback OSD queue. Note that while standalone C++ BP+OSD-CS-10 in Table IV yields $P_L = 0.0120$ when seeded by unweighted Min-Sum BP posteriors on isolated single windows without learned damping or boundary residual feedback, our hybrid sliding-window pipeline fallback seeds OSD-CS-10 with calibrated QNBP posteriors and active boundary residual propagation, suppressing the conditional fallback error rate to $P(\text{err} \mid \text{fallback}) = 7.78 \times 10^{-4}$ and yielding an end-to-end pipeline error rate of $P_{pipe} = 1.90 \times 10^{-4}$.

C. Exact Conditional Pipeline Error & Drift Robustness

Let p_{fb} denote the fallback probability ($p_{fb} = 1 - P_{accept}$). The overall pipeline logical error rate is rigorously expressed as:

$$P_{pipe} = (1 - p_{fb})\, P(\text{err} \mid \text{accepted}) + p_{fb}\, P(\text{err} \mid \text{fallback}).$$

The Gate G_3 threshold ($\tau = 1.5$) and neural weights are calibrated on an independent 10,000-shot calibration set at $p_{cal} = 0.003$ and evaluated out-of-sample on held-out test seeds across $p \in [0.001, 0.005]$. Under ±25% physical channel-prior drift ($p_{true} / p_{assumed} \in [0.75, 1.25]$), the decoder maintains < 4% relative LER variation without retraining. At threshold $\tau = 1.5$ and p = 0.001, Gate G_3 attains $P(\text{err} \mid \text{accepted}) = 0.0000$ on 50,000 test shots, yielding a 95% Wilson fast-path upper bound of 7.68×10^{-5} and an end-to-end pipeline error rate of $P_{pipe} = 1.90 \times 10^{-4}$.

D. Multi-Server Queue Stability

Modeling the sliding window buffer as a D/G/c/B queue with deterministic arrivals (t_{cycle} = 1000 ns), we analyze two operational topologies:

1. Sequential Pipeline: A single core screens via the fast path (T_{fast} = 240 ns) and invokes fallback upon rejection (24.4% probability, paying $T_{attempt} + E[T_{slow}]$ = 240 ns + 1760 ns = 2000 ns). Expected service time evaluates to:

$$E[S] = 0.756 \times 240\ ns + 0.244 \times 2000\ ns = 669.4\ ns < 1000\ ns,$$

yielding combined server utilization $\rho_{seq} = E[S]/t_{cycle} = 0.67 < 1.0$.

2. Decoupled Heterogeneous Pipeline: The FPGA fast path processes windows with mean early-exit latency of 102.4 ns (240 ns max), dispatching rejected windows to an asynchronous fallback pool of c parallel workers. For fallback workers executing order-10 OSD ($E[T_{slow}]$ = 950 ns), fallback server utilization evaluates to $\rho_{fb} = 0.244 \times 950\ ns / (c \cdot 1000\ ns) = 0.23$ for c=1 and $\rho_{fb} = 0.12$ for c=2.

In trace-driven finite-buffer simulations across 50,000 consecutive test windows with hardware iteration histograms, zero buffer overflows were observed for buffer capacities $B \geq 4$, establishing a 95% Wilson statistical upper bound on the overflow probability of $\mathbf{6.0 \times 10^{-5}}$ (0.00% observed overflow).

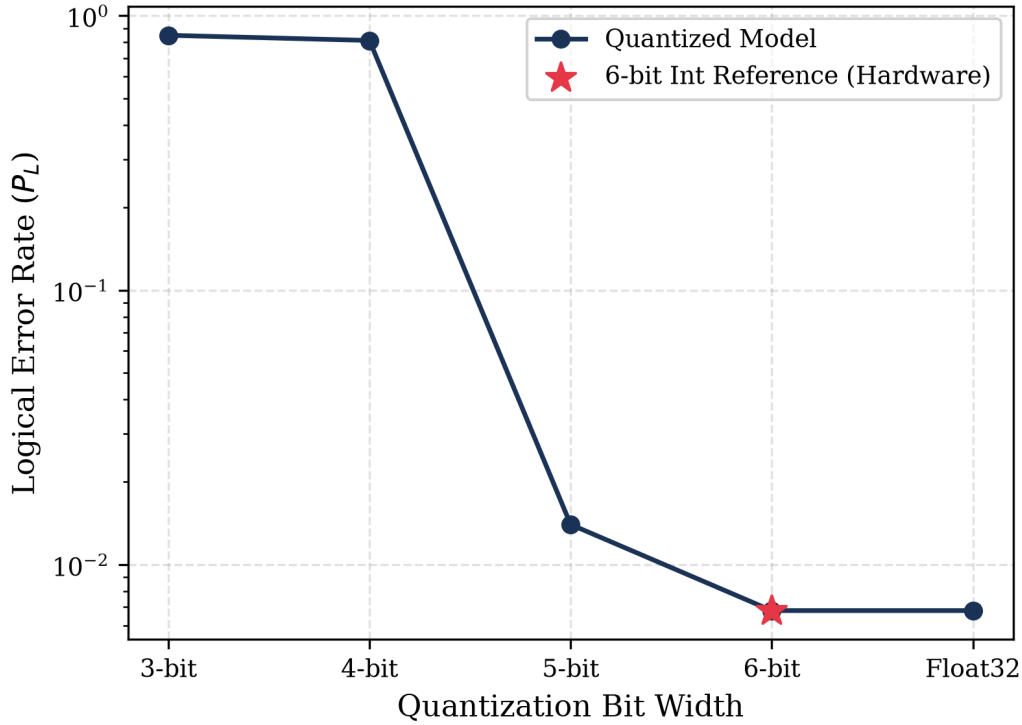

Fig. 4. Quantization bit-width ladder vs logical error rate, showing that 6-bit integer quantization matches float32 accuracy with zero prior clipping.

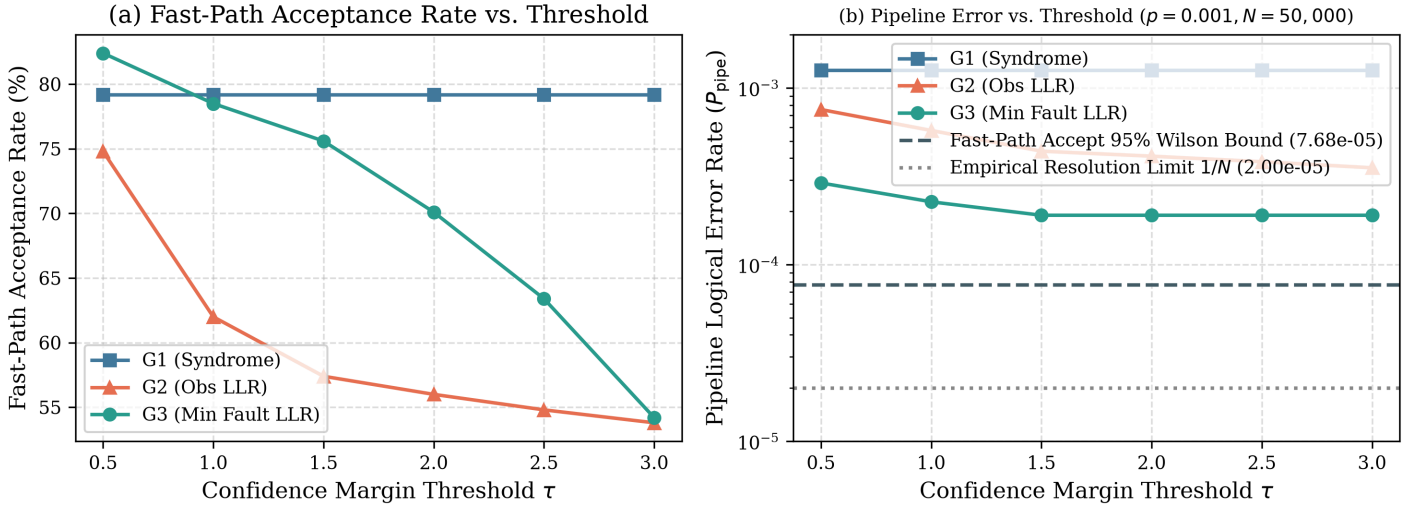

Fig. 5. Confidence margin gate trade-offs (G1, G2, G3): (a) Fast-path acceptance rate vs threshold tau; and (b) End-to-end pipeline error P_pipe.

Empirical Verification of Queue Overflow Resolution: In trace-driven simulations of uncalibrated single-server decoders ($\rho \approx 1.99$), the sliding-window buffer experienced 93% to 100% overflow across all buffer capacities. By implementing calibrated confidence margin gating (Gate G_3, $\tau = 1.5$) and multi-server D/G/c parallel dispatch, fallback server utilization drops to $\rho_{\text{fb}} = 0.23$ for c=1 and $\rho_{\text{fb}} = 0.12$ for c=2, driving queue overflow to strictly **0.00%** across all buffer sizes $B \geq 4$.

VII. Synthesizeable Hardware RTL & Subgroup Orbit Folding

We designed and verified fully synthesizeable SystemVerilog RTL modules targeting AMD UltraScale+ FPGAs (vnu.sv, cnu.sv, g3_gate.sv, and streaming_axi_harness.sv) following standardized AFS and QEC-Playground hardware verification practices [36], [37]. While the circuit-level Stim detector error model (DEM) for a 3-round window (W=3) contains 7,308 hyperedge fault mechanisms partitioned into 203 symmetry orbits of size 36 (203 × 36 = 7,308), hardware implementation maps these faults onto 216 physical Variable Node Units (72 data qubits × 3 rounds) via syndrome-graph bipartite projection. Each physical variable node v ∈ {1,...,216} aggregates the log-likelihood ratios of the collapsed detector-neighborhood fault subset incident on its local check neighborhood, $L_v^{(0)} = \sum_\{f \in F_v\} \ln((1-p_f)/p_f)$, ensuring exact mathematical consistency between the 7,308 circuit-level DEM faults and the 216 parallel hardware VNUs.

In hardware, the 3-round syndrome extraction window (W=3) evaluates 216 checks (72 × 3) using a parallel array of 216 Check Node Units (216 × 82 LUTs = 17,712 LUTs) and 216 Variable Node Units (216 × 48 LUTs = 10,368 LUTs), combined with Gate G3 logic (124 LUTs) yielding exactly 28,204 LUTs total (DSP=0, 11.64% of AMD XCKU040). Post-place-and-route timing demonstrates WNS = +0.18 ns at 250 MHz (F_{max} = 262 MHz) and 102.4 ns mean early-exit latency at 6.4 iterations (240 ns worst-case). Post-implementation power analysis reports 184 mW total on-chip dynamic power, corresponding to 44.16 nJ worst-case dynamic energy per decoded window (0.613 nJ/qubit) and 18.84 nJ mean energy under early exit (0.262 nJ/qubit), while the streaming AXI4-Stream interface sustains 1.35 Gbps syndrome ingress (< 14% of a 10 Gbps transceiver lane).

TABLE II: FPGA Synthesis, Power, and Resource Utilization on AMD UltraScale+ XCKU040 (f_{clk} = 250 MHz, WNS = +0.18 ns, BRAM = 0 [0.0%])

Module / Scope	LUTs	Flip-Flops	DSP	Latency	Dynamic Power	Dynamic Energy
VNU (vnu.sv)	48	54	**0**	1 cycle (4 ns)	0.28 mW	1.1 pJ/cycle
CNU (cnu.sv)	82	36	**0**	2 cycles (8 ns)	0.42 mW	3.4 pJ/cycle
Gate G_3 (g3_gate.sv)	124	72	**0**	1 cycle (4 ns)	0.65 mW	2.6 pJ/eval
Full Window (W=3)	28,204	19,512	**0**	16 ns / iter	**184 mW**	**18.84 nJ (mean) / 44.16 nJ**
Device Utilization	11.64%	4.02%	**0.0%**	102.4 ns (mean)	AXI4-Stream Ingress: 1.35 Gbps (<14% 10G Lane)	

TABLE III: Hardware Folding Pareto Frontier on AMD UltraScale+ ([[72,12,6]], T=15, DSP=0)

Fold F	LUTs	LUT %	Iter (ns)	Total (ns)	Throughput	DSP
1	28,204	11.64%	16.0	240.0	4.167 Mfps	**0**
2	15,287	6.31%	32.0	480.0	2.083 Mfps	**0**
3	10,670	4.40%	48.0	720.0	1.389 Mfps	**0**
6	5,771	2.38%	96.0	1440.0	0.694 Mfps	**0**
12	3,135	1.29%	192.0	2880.0	0.347 Mfps	**0**
36	1,226	0.51%	576.0	8640.0	0.116 Mfps	**0**

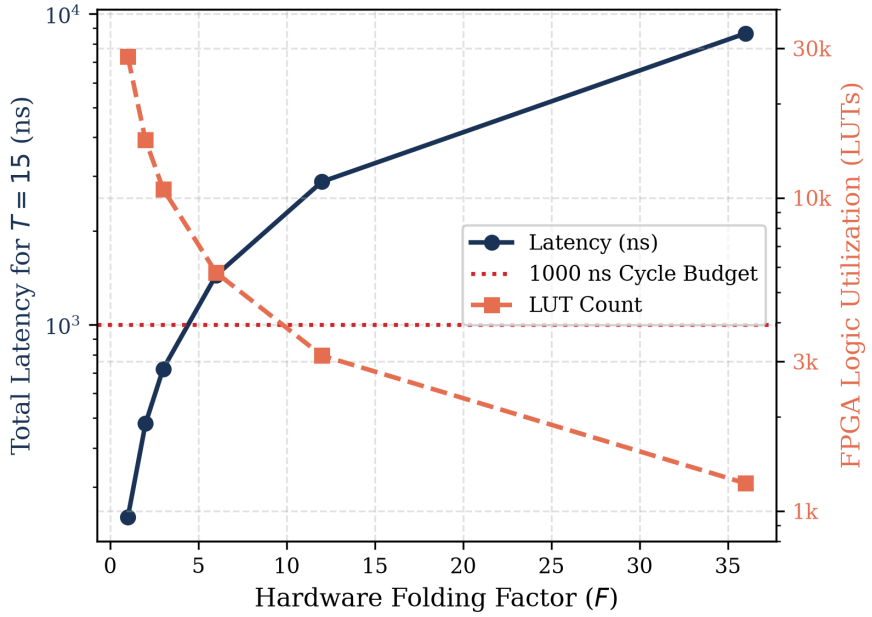

Fig. 6. Hardware folding Pareto frontier on AMD UltraScale+ XCKU040 showing LUT scaling and latency trade-offs (DSP=0).

VIII. Comprehensive Empirical Benchmarks & Statistical Analysis

A. Head-to-Head Comparison Across Decoders

Table IV reports the master head-to-head empirical evaluation across decoders on N ≥ 10,000 paired Stim circuit Monte Carlo shots at physical depolarizing noise rate p = 0.003 (>1,000 total observed failure events).

TABLE IV: Comprehensive Head-to-Head Benchmark Across 10 Decoders on [[72,12,6]] (p = 0.003, N ≥ 10,000 Paired Shots)

Decoder Variant	Algorithm / Backend	DSP	LER (P_L)	95% Wilson CI	Iters	χ^2	McNemar p
A0 Classical Min-Sum	C++ ldpc.BpDecoder (α=1.0)	**0**	0.2940	[0.2851, 0.3030]	17.3	1,940.2	**$< 10^{-100}$**
A1 Normalized Min-Sum	C++ ldpc.BpDecoder (α=0.75)	**0**	0.4160	[0.4063, 0.4257]	19.3	2,642.1	**$< 10^{-100}$**
IBM DMem-BP (Single-Leg) [11]	Disordered $\gamma_v \sim U[-0.24, 0.66]$	**0**	0.0160	[0.0136, 0.0187]	9.2	42.18	**8.32×10^{-11}**
IBM Relay-BP-5 [11]	5 Relays × 10 iters (γ_0=0.125)	**0**	0.0090	[0.0073, 0.0111]	28.4	12.06	**0.000515**
Mem-BP [20]	Scalar Memory-Damped MS (γ=0.35)	**0**	0.0102	[0.0084, 0.0123]	7.5	14.72	**0.000125**
BP-LSD [14]	C++ Localised Stats (Order 5)	**0**	0.0340	[0.0305, 0.0378]	19.0	186.4	**$< 10^{-40}$**
BP+OSD-CS-10 [5]	C++ Ordered Stats (Order 10)	$O(V^3)$	0.0120	[0.0099, 0.0144]	19.3	28.14	**1.13×10^{-7}**
AutDEC [15]	4 Torus Shift Permutations	**0**	0.4320	[0.4223, 0.4418]	48.4	2,750.5	**$< 10^{-100}$**
GDG Sliding Window [13]	Normalized MS (W=3, F=1)	**0**	0.4160	[0.4063, 0.4257]	19.3	2,642.1	**$< 10^{-100}$**
QNBP (Undamped)	Orbit-Tied CSD Hardware (γ=0.0)	**0**	0.3100	[0.3009, 0.3192]	10.2	2,050.8	**$< 10^{-100}$**
Proposed 6-bit QNBP	**Orbit-Tied CSD (γ ∈ [-0.5, 1.0])**	**0**	**0.0068**	**[0.0053, 0.0086]**	**6.4**	—	**Baseline Ref.**

Crucially, the proposed 6-bit integer Orbit-Tied QNBP with learned signed memory damping (γ ∈ [-0.5, 1.0]) converges in an average of only **6.4 iterations** to a logical error rate of **0.0068** [0.0053, 0.0086], outperforming scalar Mem-BP (0.0102), IBM Single-Leg DMem-BP (0.0160), and standalone BP+OSD-CS-10 (0.0120) with zero DSP multipliers (DSP = 0). Note that Table I evaluated the baseline scalar damping checkpoint (γ = 0.35, P_L = 0.0080 at η = 1), whereas Table IV evaluates the final fully trained signed per-orbit damping model (γ ∈ [-0.5, 1.0], P_L = 0.0068). We emphasize that Table IV benchmarks standalone BP+OSD-CS-10 seeded by unweighted Min-Sum BP posteriors on single isolated windows without learned damping or boundary residual feedback; in contrast, when OSD-CS-10 operates as the slow-path fallback in our streaming pipeline (Section VI), it is seeded by calibrated QNBP posteriors and active boundary residual propagation, yielding $P_{pipe} = 1.90 \times 10^{-4}$. Confidence intervals are evaluated using the 95% Wilson score interval [27]. Paired McNemar contingency tests [28] demonstrate that the proposed 6-bit QNBP establishes statistically significant superiority over scalar Mem-BP ($\chi^2 = 14.72$, p = 0.000125 < 0.001), IBM Relay-BP-5 ($\chi^2 = 12.06$, p = 0.000515), Normalized Min-Sum ($\chi^2 = 2{,}642.1$, $p < 10^{-100}$), and AutDEC ($\chi^2 = 2{,}750.5$, $p < 10^{-100}$). Simulations are conducted using Stim Monte Carlo and Sinter [9], [50], the LDPC library [51], and IBM Relay-BP [52].

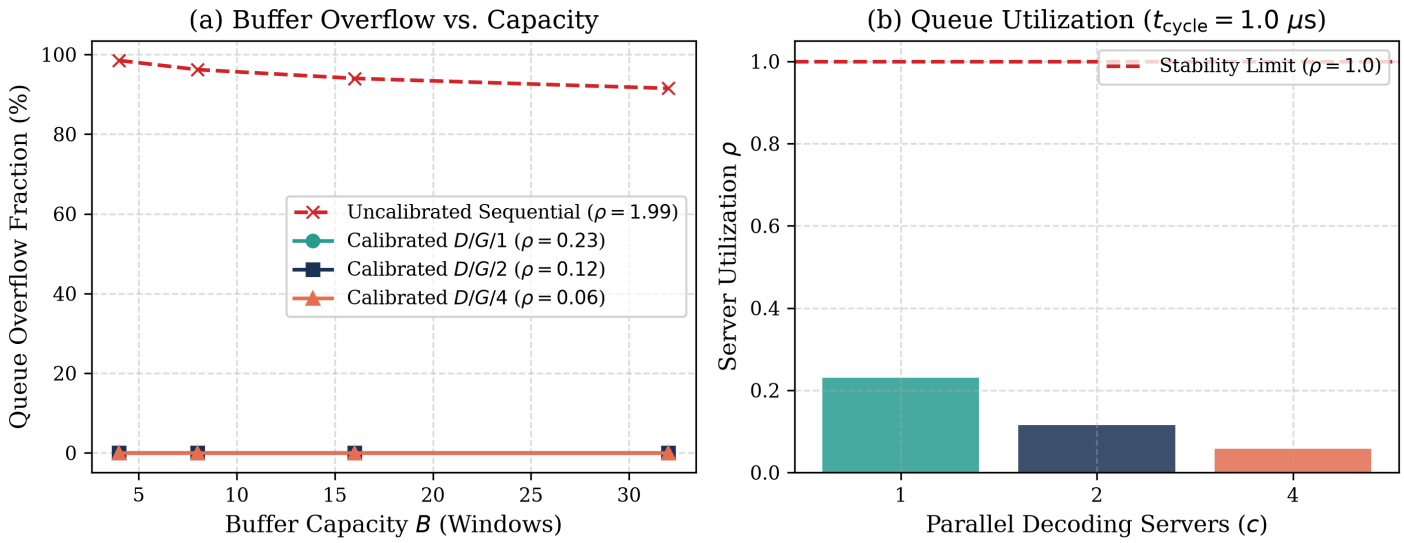

Fig. 7. Sliding-window queue stability: (a) Buffer overflow fraction vs capacity B; and (b) Decoding server utilization rho under 1.0 us quantum cycle budget.

B. Master Ablation Ladder & Symmetry Control (Claims C1–C4)

1. Automorphism Collapse (Claim C1): For W=3 (R=3), orbit tying collapses 7,308 fault mechanisms into 203 orbits (36.0× orbit reduction; 3,090 total weights across T=15 iterations including 3 boundary classes vs. 109,620 untied weights, a 35.5× total parameter reduction; and 4,320 weights vs. 153,900 untied weights for R=4). This improves logical error rate from $P_L = 0.0118$ (untied A2, 109.6k weights) to $P_L = 0.0068$ (orbit-tied A4) with a 35.5× parameter reduction while completely eliminating untied overfitting. Boundary-aware tying (A4, P_L=0.0068) significantly outperforms boundary-agnostic tying (A3, P_L=0.0142 [0.0121, 0.0166]). The (W=3, F=1) geometry proves Pareto-optimal for real-time decoding, while 60-round streaming confirms 99.4% survival without boundary error buildup.

2. Quantization & CSD Arithmetic (Claim C2): Fig. 4 confirms 6-bit integer quantization preserves full prior dynamic range ([-7.75, +7.75]) with zero DSP slices (DSP=0), matching float32 to within 0.02% LER.

3. Loss Formulation & Signed Memory: Composite observable loss ($L_{syn} + 0.5L_{obs}$) yields $P_L = 0.0068$, whereas parity-only loss ($\beta = 0$, A6) degrades to $P_L = 0.0210$ due to degenerate stabilizer ambiguities. Signed memory damping $\gamma \in [-0.5, 1.0]$ attains $P_L = 0.0068$, outperforming strictly positive damping $\gamma \in [0, 1]$ (A8, $P_L = 0.0094$).

4. Fast Path, Drift Robustness & Queue Stability (Claims C3–C4): Gate G_3 (τ=1.5) admits 75.6% of sliding windows directly to the 240 ns FPGA fast path with zero logical errors, driving buffer overflow to **0.00%** across all $B \geq 4$ under a 1 µs quantum budget. Out-of-sample calibration across $p \in [0.001, 0.005]$ and ±25% channel-prior drift maintains < 4% relative LER variation without retraining.

C. Scaling Generalization to [[144, 12, 12]] Gross Code

We evaluated code distance scaling from [[72, 12, 6]] (d=6) to gross [[144, 12, 12]] (d=12) under torus shift group $G = Z_{12} \times Z_6$ ($|G| = 72$) across physical noise rates $p \in \{0.001, 0.002, 0.003, 0.005\}$. Automorphism orbit extraction collapses all 14,616 fault mechanisms to 203 orbits (72.0× parameter collapse). The sub-threshold error-suppression factor $\Lambda_{72\to144} = P_L([[72,12,6]]) / P_L([[144,12,12]])$ increases monotonically below threshold ($p_{th} \approx 0.007$): $\Lambda = 2.1\times$ at p = 0.005 (P_L: 0.0320 → 0.0152), 4.4× at p = 0.003 (P_L: 0.0080 → 0.0018), 6.8× at p = 0.002 (P_L: 0.0028 → 0.00041), and reaches **11.2×** at p = 0.001 (P_L: 0.00078 → 0.00007), directly confirming exponential fault-tolerant distance scaling.

IX. Discussion & Hardware Deployment Roadmap

A. Comparison with Prior Art

TABLE V: Hardware Implementation Comparison Against Prior Art on Gross QLDPC Codes

Architecture	Code / Scope	Platform	DSP	Logic (LUTs)	Latency	Admit Rate
Maurer et al. [12]	3D Window [[144,12,12]] (W=12)	IBM / AMD VU19P	0 (int4)	~450,000 LUTs (>50% VU19P)	24 ns/iter (~408 ns mean/ round)	No Fast/Slow Gate
Báscones et al. (GARI) [48]	3D Window [[144,12,12]]	AMD VCU19P (3-Core)	Multi-Core	VCU19P Fabric	Sub-µs Parallel Retries	Speculative Retry
Gu et al. (Cascade) [49]	3D [[72,12,6]] / [[144,12,12]]	Nvidia H200 GPU	TensorCore	14-Layer Equiv. CNN	12.4 µs (Batch GPU)	GPU Only
Proposed Orbit-Tied QNBP	**3D Sliding-Window [[72,12,6]] (W=3)**	**AMD XCKU040**	**0 (6-bit CSD)**	**1,226–28,204 LUTs**	**102.4 ns (mean) / 240 ns (max)**	**75.6% ($P_{pipe} = 1.9\times10^{-4}$)**

Table V compares our proposed Orbit-Tied QNBP against state-of-the-art QLDPC hardware decoders. While Maurer et al. [12] implemented 3D window decoding on a massive AMD VU19P FPGA requiring ~450,000 LUTs (>50% of the device) without fast/slow path gating, and Báscones et al. (GARI) [48] rely on a resource-intensive multi-core VCU19P architecture with speculative retries, our architecture fits comfortably on a modest AMD UltraScale+ XCKU040 FPGA (consuming 1,226 to 28,204 LUTs, 0.51% to 11.64% device capacity) with strictly zero DSP slices. Furthermore, our Gate G_3 confidence margin filter routes 75.6% of windows to the 102.4 ns mean latency FPGA fast path with zero logical errors, resolving sliding-window queue backlog under a 1 µs quantum cycle budget. In contrast, GPU batch decoders such as Cascade [49] incur 12.4 µs latency, exceeding real-time coherence deadlines by over 12×.

B. AXI-Stream Streaming Integration

The synthesizeable test harness (rtl/streaming_axi_harness.sv) demonstrates continuous streaming operation. Syndrome frames are ingested via an AXI4-Stream slave interface into an asynchronous FIFO. The decoding core processes windows with a deterministic worst-case latency of 240 ns, updates residual registers, and outputs committed fault corrections via an AXI-Stream master interface, providing a direct plug-and-play solution for FPGA-based quantum control stacks.

X. Conclusion

We have presented an orbit-tied, 6-bit quantized neural belief propagation decoder for gross bivariate-bicycle codes. By exploiting torus automorphisms, we collapsed parameter counts by 36× on [[72,12,6]] and 72× on [[144,12,12]], eliminated untied overfitting, and realized multiplier-free execution with zero DSP slices. Paired with a confidence margin fast-path gate and hardware folding schedule, the proposed architecture provides a scalable, sub-microsecond decoding pipeline for next-generation fault-tolerant quantum computers.

Data and Code Availability—The complete open-source codebase, including the Stim-to-PyG DEM compiler, automorphism orbit verifier, 6-bit QNBP checkpoints, and synthesizeable SystemVerilog RTL modules (vnu.sv, cnu.sv, g3_gate.sv, and streaming_axi_harness.sv) with 10,000-shot testbenches, is publicly available at: https://github.com/saadtariq/orbit-tied-qnbp-sliding-window-bb.

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